

PRAGATI KORE

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EDUCATION

Baylor College of Medicine	Doctor of Philosophy, Genetics & Genomics Bioinformatics, Genomics and Systems Biology (BiGSB) Track Clinical Translational Research Certificate of Added Qualifications (CTR-CAQ) Overall GPA: 4.00	August 2022 - Present
The University of Texas at Austin	Bachelor of Science, Biochemistry Certificate: Elements of Computing Overall GPA: 3.48	August 2016 - May 2020

HONORS, GRANTS & AWARDS

- Leena Peltonen School of Human Genomics July 2026
Selected as one of twenty early-career researchers nationwide to participate in this workshop
- Honorable Mention, Gertrude M. Cox Scholarship, American Statistical Association April 2026
- 2026 Weintraub Graduate Student Award Nominee November 2025
Selected by the BCM Genetics Program Executive Committee
- UCLA Emerging Genomic Scientist Scholar April 2025
Selected as one of five early-career researchers nationwide to present at UCLA's 2025 Emerging Genomic Scientist Symposium
- NIH T32 GM136554-05 Training Grant August 2024 - July 2025
- Advanced Gene Mapping Course with a Travel Grant at The Rockefeller University, New York April 2024
- NIH T32 GM139534-03 Training Grant August 2023 - July 2024
- Professor John J. Trentin Scholarship Award October 2023
- NIH T32 GM139534-02 Training Grant August 2022 - July 2023
- Undergraduate Research Fellowship October 2019
Proposal Title: "Exploring the Variance Structure of a Socially Sensitive Transcriptome"
Awarding Institution: The University of Texas at Austin
Amount: \$1000
- University Honors Fall 2016, Fall 2017, Spring 2018
- Women in Natural Sciences Residential Program (WINS) August 2016 – May 2020
WINS Mentor (Fall 2017-Spring 2019) - advise and offer counsel to a current underclassman in WINS

GRADUATE CAREER

Graduate Research Assistant (PhD Dissertation Work)

Advisor: Dr. Elizabeth Atkinson, Dept. of Genetics and Genomics, BCM, Houston, TX, USA

January 2023 – Anticipated December 2026

Thesis Project: Advancing Genomic Equity through Local Ancestry Inference

Part 1: Improved Allele Frequencies in gnomAD through Local Ancestry Inference

- Collaboration with the Broad Institute of MIT and Harvard as a visiting researcher under Dr. Mark Daly
- **Kore, P.***, Wilson, M.W.*, Tiao, G. et al. Improved allele frequencies in gnomAD through local ancestry inference. Nature Communications 16, 8734 (2025). <https://doi.org/10.1038/s41467-025-63340-2>
 - Received wide media attention
 - Invited talk, *BCM Molecular and Human Genetics Departmental Retreat - January 2026*
 - Invited talk, *Cassyni Research Seminars - December 2025*. <https://doi.org/10.52843/cassyni.gtwbzw>

- Invited talk, *Baylor College of Medicine-Baylor Genetics Collaborative Meetings* - June 2025
- Invited talk, *UCLA Emerging Genomic Scientist Symposium* – April 2025
- Invited talk, *gnomAD council at the Broad Institute of MIT and Harvard* – Sept. 2024

- This work represents a significant advance in the ancestral granularity of reference resources, introducing local ancestry-informed allele frequencies in gnomAD for two major admixed populations: the African/African-American and Admixed American inferred genetic ancestry groups.
- While previous releases of gnomAD allele frequencies have provided valuable insights into human genetic diversity, they are based on aggregate estimates that obscure genetic differences across ancestral populations, particularly for recently admixed groups.
- By refining allele frequency estimates with local ancestry deconvolution, our work offers a major improvement in understanding population-enriched genetic variation, particularly crucial for the accurate interpretation of variants in understudied populations.
- This release marks the first time local ancestry has been systematically incorporated into a globally-representative resource like the Genome Aggregation Database (gnomAD), setting a new standard for genomic inclusion.

Part 2: Improving Rare Variant Discovery in Admixed Populations

- Despite recent advances in understanding the genetics of complex diseases, there remain major limitations in our knowledge of their genetic architecture in diverse and minority populations due to the lack of statistical genomics tools tailored for diverse groups. My project focuses on the development of a novel rare variant association study framework that leverages local ancestry inference to deconvolve the ancestral origin of each region of an individual's genome, leading to the improved ability to detect risk genes in mixed-ancestry populations. Our novel tool will facilitate the inclusion of diverse populations in large-scale rare variant analyses and improve our understanding of the genetic architecture of complex traits across different ancestries, leading to improved personalized medicine in underserved populations.

CTR-CAQ Capstone Project: Advancing Risk Stratification in Wolff-Parkinson-White Syndrome: Polygenic Risk Score for Atrial Fibrillation

- Collaboration with Dr. Seema Lalani and Dr. Christina Miyake
- Wolff-Parkinson-White (WPW) syndrome is a cardiac condition involving an accessory electrical pathway and an elevated risk for atrial fibrillation (AF). Additionally, WPW syndrome is known to predispose individuals to life-threatening arrhythmias such as AF, but the genetic relationship between WPW and AF is poorly understood. Thus, I developed a polygenic risk score (PRS) for AF to classify WPW cases versus population controls from the 1000 Genomes Project. Using GWAS summary statistics and whole-exome sequencing data from 305 individuals—including trios, singletons, and families—I constructed and validated a PRS model that distinguished WPW cases from controls. The goal was to support clinical decision-making by improving genetic classification of WPW status. Ultimately, this work advances precision medicine in cardiology by uncovering genetic contributions to arrhythmic risk in WPW and improving individualized care strategies.
- Invited talk, *CTR-CAQ Annual Retreat* - Sept. 2025

Lab Rotation

Advisor: Dr. Philip Lupo, Dept. of Genetics and Genomics, BCM, Houston, TX, USA

March 2023 – May 2023

- Topic: Determine the extent to which genetic ancestry is correlated with acute lymphoblastic leukemia (ALL) subtypes by using ADMIXTURE software
- **Kore, P.***, Geris, J.M.*, Leon-Camarena, M. et al. Interplay between Genetic Ancestry, Self-reported Race and Ethnicity, and Clinical Factors in Pediatric Acute Lymphoblastic Leukemia: A REDIAL Consortium Report. *Cancer Epidemiol Biomarkers Prev* 2026; <https://doi.org/10.1158/1055-9965.EPI-25-1438>

Lab Rotation

Advisor: Dr. Chao Cheng, Dept. of Genetics and Genomics, BCM, Houston, TX, USA

October 2020 – December 2022

- Topic: Optimization of automatic spatial annotation using adenocarcinoma samples
- Based on the annotation from the pathologist, I defined gene signature for each anatomic types dependent on Voxel-expression data
- Applied gene signature to other tissue slides to determine regions of the same anatomic type

Lab Rotation

Advisor: Dr. Nick Tran, Dept. of Genetics and Genomics, BCM, Houston, TX, USA

August 2022 – October 2022

- Topic: Development of SARM1 deletion strategies to block wallerian degeneration and protect retinal ganglion cells (RGCs) in the retina
- Designed CRISPR guides, performed in situ DNA-HCR, and conducted cloning for SARM1 to eventually package adeno-associated virus (AAV) vectors to co-express the gRNA and Cas9
- Observed and conducted certain cell dissociations, retinal microdissections, tissue processing and staining, and learned basics of plotting and analyzing scRNA-seq data

PROFESSIONAL EXPERIENCE

BioMarin Pharmaceutical Inc. – *Genomics Intern (Remote); San Rafael, California*

June 2025 – September 2025

- Led large-scale genomic analyses to identify diseases with more diagnosed patients in our sponsored testing program than expected, leading to an existing under-recognized commercial opportunity
- Identified genes with excess variants of uncertain significance (VUS) in tested patients, where improved classification could increase commercial opportunity through boosted patient uptake
- Applied unsupervised clustering to reveal functional gene relationships and prioritize candidate genes for skeletal disorders
- Performed ROI analysis for WGS implementation in skeletal dysplasia diagnostics, delivering cost-benefit models and clinical utility metrics to guide strategic investment decisions
- Developed reproducible R pipelines and documentation to streamline analyses
- Delivered 3 oral presentations to the Genomics and Skeletal Discovery teams; Collaborated with clinical geneticists, bioinformatics, and discovery teams to turn results into prioritized next steps

Icahn School of Medicine at Mount Sinai (Sebra Lab) – *Associate Researcher; New York, New York*

September 2020 – June 2022

- Conducted R&D and for-service genomics projects using 10x genomics technology with the Sinai community and beyond
- Assisted in the collection, analysis and review of various experimental data for publication and presentation
- Analyzed High-Grade Serous Ovarian Cancer data in order to identify significant patterns for further publications
- Performed cell culture techniques to assess patient endometrial tumor data
- Utilized the Beacon Optofluidic System to identify how certain cell signaling pathways react to various drug treatments

The University of Texas at Austin – *Paid Texas Institute for Discovery Education in Science (TIDES) Evaluation Fellow; Austin, Texas*

January 2020 – May 2020

- Conducted survey research, including survey design, administration and logistics
- Designed, collected, analyzed, created reports, and presented data and information - reports included tables, charts, dashboards, and executive summaries
- Aimed to expand the capacity of the College of Natural Sciences by conducting education research and evaluation to assess the usefulness and impact of education programs and initiatives

The University of Texas at Austin – *Paid Undergraduate Teaching Assistant (UGTA) for Genetics; Austin, Texas*

August 2019 – May 2020

- Attended and facilitated the class by assisting with class participation activities
- Taught discussion sections for 90 students outside of lecture to help students understand course subject-matter by creating and providing additional materials
- Participated in weekly meetings with the other four Undergraduate Teaching Assistants and the professor about ways to improve the class environment based on a review of the work done in the previous week
- Printed, proctored and graded quizzes/exams in a timely manner according to guidelines set by the professor
- Held weekly office hours to give students an extra chance to ask questions and get help with the material

Merck & Co. Inc. – *MRL Pharmacokinetics, Pharmacodynamics, & Drug Metabolism Bioanalytics Paid Intern; Kenilworth, New Jersey*
May 2019 – August 2019

- Department is responsible for assessing immunogenicity for the vaccines and biologics pipeline, at the preclinical and early clinical stage
- Worked with Anti-drug Antibody assays for optimization and discovered the drug tolerance of the ligand-binding assay
- Utilized the new ADA DOE workflow to perform a “look back” for three prior ADA developments
- Experience with operating the Tecan liquid handling system
- Participated in the evaluation of a novel ADA development workflow that encompasses the design of experiment (DOE) and lab automation components
- Understood the importance of immunogenicity assessment for biologics drug candidates and familiarized with DOE concepts
- Interpreted data by performing R scripting via various queries
- Utilized Design Expert software for data analysis and acquired data for manuscript publication
- Oral and poster presentation of the summer project was given to the leadership team

Asuragen – *Summer Research Paid Intern; Austin, Texas*
May 2018 – August 2018

- Headed a project regarding a rare inherited disease (Friedreich's ataxia) that causes progressive nervous system damage and movement problems
- Molecular techniques practiced include PCR, ddPCR, CE, Quant-IT kit, and many more
- Purpose of project was to invent a kit that allows clinics and other scientists to identify if a patient has this disease or if they are a carrier or not; giving them the exact molecular information correlating with their results
- Learned how to create experiments independently as well as design, order, and test primers
- Oral and poster presentation of the summer project given to the leadership team

UNDERGRADUATE RESEARCH EXPERIENCE

The Hofmann Lab – *Department of Integrative Biology – Undergraduate Bioinformatics Intern; Austin, Texas*
September 2018 – December 2019

- Goal of the research was to identify the genes and gene modules involved in response to different environmental factors
- Analyzed the transcriptomes of the brains of animals subjected to various environmental perturbations using bioinformatic and statistical approaches in the programming environment RStudio
- Examined the behavior of treated and control animals in a standard sociability assay, then measured circulating hormones and other physiological parameters
- Used weighted gene co-expression network analysis (WGCNA) to identify modules of co-regulated genes that might be associated with any EDC-dependent effects on behavior
- Presented a poster at the TACCSTER 2019 Conference at the J.J. Pickle Research Campus of UT Austin
- Contributed to a publication (See below)
- Self-written proposal, "Exploring the Variance Structure of a Socially Sensitive Transcriptome", was selected as an awardee for the Undergraduate Research Fellowship and was awarded \$1000

The Jha Lab - Department of Integrative Biology - Undergraduate Research Intern; Austin, Texas

May 2017 – May 2018

- Purpose of project was to examine how regularly occurring extreme weather events affect annual pollen flow patterns in tropical trees
- Performed microscopy and molecular processing of plant population genetic samples using microsatellite markers
- Molecular techniques practiced include processing and analyzing DNA extraction from thousands of samples and multiplex PCR assays
- Contributed to a publication (See below)

PUBLICATIONS

- **Kore, P.***, Geris, J.M.*, Leon-Camarena, M. et al. Interplay between Genetic Ancestry, Self-reported Race and Ethnicity, and Clinical Factors in Pediatric Acute Lymphoblastic Leukemia: A REDIAL Consortium Report. *Cancer Epidemiol Biomarkers Prev* 2026. <https://doi.org/10.1158/1055-9965.EPI-25-1438>
- **Kore, P.***, Wilson, M.W.*, Tiao, G. *et al.* Improved allele frequencies in gnomAD through local ancestry inference. *Nature Communications* 16, 8734 (2025). <https://doi.org/10.1038/s41467-025-63340-2>
- Laitman BM, Charytonowicz D, Zhu AJ, Lynch L, Varelas EA, Burton M, Andreou C, **Kore P**, Beaumont KG, Kirke DN, Genden EM, Courey MS. High resolution profiling of human vocal fold cellular landscapes with single-nuclei RNA sequencing, *Laryngoscope*, 2024. <https://doi.org/10.1002/lary.31334>
- Morgan E Hernandez Scudder, Rebecca L Young, Lindsay M Thompson, **Pragati Kore**, David Crews, Hans A Hofmann, Andrea C Gore, EDCs Reorganize Brain-Behavior Phenotypic Relationships in Rats, *Journal of the Endocrine Society*, Volume 5, Issue 5, May 2021, bvab021, <https://doi.org/10.1210/jendso/bvab021>
- Wisath Sae-Lee, Luisa L Scott, Lotti Brose, Aliyah J Encarnacion, Ted Shi, **Pragati Kore**, Lashaun O Oyibo, Congxi Ye, Susan K Rozmiarek, Jonathan T Pierce, APP-Induced Patterned Neurodegeneration Is Exacerbated by APOE4 in *Caenorhabditis elegans*, *G3 Genes/Genomes/Genetics*, Volume 10, Issue 8, 1 August 2020, Pages 2851–2861, <https://doi.org/10.1534/g3.120.401486>
- Megan O’Connell, Antonio R. Castilla, Hilario Espinosa, **Pragati Kore**, Apoorva Magadi, Alonso Santos-Murgas, Shalene Jha (*manuscript submitted for publication*). Landscape Genetic Variation Enhances Plant Reproductive Resilience to Climate Extremes, November 2024.

BLOG POSTS/RESEARCH COMMUNICATION

- **Kore, P.** “Why Local Ancestry Matters.” Baylor College of Medicine Blog, 2026.
- **Kore, P.**, Wilson, M., et al. “Local Ancestry Inference for African/African American Samples in gnomAD.” Genome Aggregation Database (gnomAD) Blog, 2024.

CONFERENCE ABSTRACTS & POSTERS

- **Pragati Kore**, Morgan E. Hernandez, Lindsay M. Thompson, David Crews, Rebecca L. Young, Hans A. Hofmann, Andrea C. Gore (2019) Endocrine-Disrupting Chemicals Affect Social Behavior and Brain Gene Expression Differently in Male and Female Rats. TACCSTER 2019 Conference
- **Pragati Kore**, Lai Yeung, Shara Dellatore (2019) Anti-Drug Antibody (ADA) Assay Development. Merck Summer Research Poster Symposium at Merck & Co. Inc. (Headquarters)
- **Pragati Kore**, Amy Marshall, Hannah Welch, Elise Connor, Moriah Sandy¹, Christine V. Hawkes², ¹Research Educator, ²Principal Investigator (2018) Analysis of Endophytic Fungi Under Varying Osmotic Stress in Pure and Competing Cultures. Undergraduate Research Forum at The University of Texas at Austin
- **Pragati Kore***, Jennifer Geris*, Maria Leon-Camarena, Maria Castellanos, Olga Taylor, Veronica Garcia-Morales, Austin Brown, Pagna Sok, Van Huynh, Kathleen Ludwig, Laura Blesse, Sandi Pruitt, Amy Hughes, Kevin Tien, Kenneth Heym, Timothy Griffin, Rodrigo Erana, Juan Carlos Bernini, Monica Gramatges, Karen Rabin, Michael Scheurer, Philip Lupo. Association of Inferred Global Genetic Ancestry with Cytogenetic Subtypes in Pediatric Acute Lymphoblastic Leukemia. American Society of Human Genetics (ASHG) Nov. 2023.

- **Pragati Kore***, Michael W. Wilson*, Grace Tiao, Katherine Chao, Phil Darnowsky, Nick Watts, Samantha M. Baxter, Jessica Honorato Mauer, Genome Aggregation Database Consortium, Heidi L. Rehm, Mark J. Daly, Konrad J. Karczewski, Elizabeth Atkinson. A Public Resource of gnomAD Local Ancestry Inference Data for Diverse Human Genomes (ASHG - Nov. 2024, Broad Institute of MIT and Harvard Retreat - Dec. 2024, Molecular and Human Genetics Retreat - Jan. 2025, Gordon Research Seminar/Conference - July 2025)
- Sarah Tucker, **Pragati Kore**, Chelsea Boyd, Carlos A. Lodeiro, Brock A. Karolcik, Tam Dan N. Pham, Bryan C. Cannon, Jeffrey J. Kim, Santiago O. Valdés, Taylor S. Howard, Wilson W. Lam, Christina Miyake. Atrial Fibrillation and High-Risk Features during Electrophysiology Study are Not Associated with Genetic Variants in a Study of 108 Pediatric Patients with Wolff-Parkinson-White Syndrome. Heart Rhythm Society April 2026.

MEMBERSHIPS

- American Society of Human Genetics (ASHG) – Trainee (2023 & 2024)

WORKSHOPS ATTENDED

- Advanced Gene Mapping Course at The Rockefeller University, New York (2024)
- 1) Race, Ancestry, Both, or Neither?: Introducing Frameworks for Population Descriptors in Genomics Research, 2) Genomic Analysis in the All of Us Researcher Workbench, 3) Behind-the-Scenes: Publications Workshop - ASHG, Washington, DC (2023)

TECHNICAL SKILLS

Biomedical Laboratory Skills: Experience with handling the autoclave for sterilization, precise pipetting, setting up and running a PCR (Polymerase Chain Reaction), running a CE (Capillary Electrophoresis), 10x genomics technology, handling a spectrometer, conducting a bacterial transformation, performing a gel electrophoresis, conducting cell assays and culture stocks, handling a Beacon Optofluidic System, working with a UV-VIS Nanodrop, preparing different buffers and solutions, utilizing analytical and pH scales, creating Petri dishes and agar gels from scratch, using the Rotary Evaporator, operating the Freedom EVO 75 and 200 base unit Tecan machine, and working with cell culture under the hood.

Coding Skills: Python, RStudio, SQL, Swift, Google Cloud Platform, Linux, Unix, Plink, Hail genomics

Research Computing/Analysis Skills: Image J, Image Pro Plus, Genemapper, Design Expert, Rosalind, Reactome

Office Software Skills: MS Word, Excel, PowerPoint

Desktop Publishing Skills: Photoshop and After Effects

LEADERSHIP EXPERIENCE AND ACTIVITIES

Saturday Morning Science Career Development Mentorship Program at BCM – *career development mentor*

October 2023 – Present

- I am matched with a mentee from a pool of 7th-12th grade students that are underrepresented in medicine
- Bimonthly meetings to advise mentee on career development opportunities based on their interests and long-term goals

CTR-CAQ Executive Steering Committee – *class representative*

October 2023 – September 2025

- I acted as a liaison between the committee and class members, ensuring effective communication and representation of multiple perspectives
- I collaborated with fellow representatives to develop and implement strategies for achieving goals held by the program

Children's Museum in Houston – *volunteer*

August 2023 – February 2025

- Facilitated hands-on exhibits, craft projects, and interactive science experiments

- Extended experiences to underserved areas of Houston, specifically to schools that might not have easy access to such opportunities.
- Helped shape a more inclusive and diverse environment for both the scientific and local community

Department of Molecular and Human Genetics at BCM – *volunteer*

August 2022 – May 2024

- Microart judge (May 2023): I helped select the top SciArt to be featured at the American Society for Microbiology (ASM) MicroArt event during ASM Microbe 2023 in Houston.
- National DNA Day volunteer (April 2023): The aim was to educate people about the latest advances in genomic research and their impact on their lives. Combining this objective with the departmental goal to foster an environment of inclusion, education, and understanding with the community, we invited local students to tour our labs and learn about Genetics and DNA

Austin Disaster Relief Network – *volunteer*

April 2020 – September 2020

- Sewed cloth masks at home using personal sewing machine to distribute to the community during the COVID-19 pandemic

SEAL (Students Expanding Austin Literacy) – *selected mentor*

September 2017 – May 2019

- Members volunteered as mentors and educators to small groups of at-risk elementary students on a weekly basis
- Assisted and tutored kids with reading
- Promoted confidence, learning, and an appreciation for community service in both our members and our children

ICA (Indian Cultural Association) – *member*

August 2016 – May 2020

- Members and neighboring organizations came together to promote culture, service, friendship, professionalism, and leadership
- Oversaw and facilitated events, hosting teams from Universities all over the country as a committee member

FRI (Freshman Research Initiative) Ambassador – *selected member*

March 2017 – May 2020

- Worked closely with the director of FRI
- Represented FRI by informing high schoolers about our experience in the program at both on and off-campus events
- Held numerous events year-round that FRI Ambassadors assisted with - such as at Explore UT, Alumni Events, the Undergraduate Research Forum, and Summer Orientation recruitment

LEAP (Leadership Education and Progress) Program – *selected member*

January 2017 - March 2017

- Program dedicated to preparing students for leadership and networking
- Developed a foundation for leadership through the Social Change Model
 - has helped me improve my communication skills as an FRI Ambassador
- Learned to engage as an ethical and inclusive leader by UT dean's and community leaders

ADDITIONAL INFORMATION

Languages: Fluent in English, Working knowledge in Hindi and Marathi, Conversational in Spanish

Hobbies: Play the flute, play basketball, enjoy cooking and traveling

Work Eligibility: Eligible to work in the U.S. with no restrictions